

Prompt Engineering Study Handbook

From Beginner to Advanced — Your Complete Reference Guide

Based on ABTalks × 60 Days of AI Prompt Engineering Roadmap

21

Chapters

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Templates

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Day Plan

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60 Days -
60 Days Claude Challenge
60 Days Claude Challenge by ABTalks
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CHAPTER 1

What Is Prompt Engineering?

1.1 Definition

Prompt engineering is the practice of designing, structuring, and refining instructions (prompts) that guide an AI language model to produce accurate, relevant, and high-quality outputs — reliably, on the first try.

1.2 Why It Matters

- Saves time by reducing back-and-forth corrections
- Unlocks capabilities most users never discover
- Essential for freelancing, consulting, product development, and enterprise AI
- Differentiates casual users from professional AI practitioners
- Applicable across writing, coding, analysis, design, education, marketing

1.3 The Quality Ladder

Level	Example Prompt	Strength / Problem
Weak	write about marketing	Vague — generic essay
Okay	write 5 marketing tips for a small bakery	Better, but lacks persona/format
Good	Act as a marketing consultant. 5 low-budget Instagram ideas.	Better. Basic format
Pro	Act as marketing consultant, 10 yrs food retail. Budget: \$1k. 5 ideas + roleplay action, outcome. Table.	5k. 5 ideas + roleplay action, outcome. Table.

■ You don't need fancy words — you need specificity. Every detail removes a guess the AI would otherwise make.

1.4 Common Beginner Mistakes

- Writing one vague line and hoping for the best
- Cramming multiple unrelated tasks into a single prompt
- Assuming the AI already knows your context
- Accepting the first draft as final
- Forgetting to specify who the output is for

CHAPTER 2

Fundamentals of Prompt Engineering

2.1 How AI Models Read a Prompt

AI models process your entire prompt as tokens (chunks of text), predicting the most statistically likely continuation. This means:

- **Order matters** — context given early shapes what comes later
- **Clarity matters** — ambiguity leads to averaged, generic outputs
- **Structure matters** — well-organised prompts produce well-organised responses

2.2 Prompt Types Overview

Type	Description	Best For
Zero-Shot	No examples; AI uses training alone	Quick tasks, factual queries
Few-Shot	1–3 examples provided in the prompt	Tone/style matching, structured outputs
Chain-of-Thought	Ask AI to reason step by step	Math, logic, complex decisions
Role-Based	Assign AI a specific expert persona	Expert-level output in any domain
XML Structured	Use structured markup tags	API workflows, agentic systems
Prompt Chaining	Break task into sequential prompts	Long docs, multi-step processes

2.3 Key Vocabulary

Term	Meaning
Token	A chunk of text (~4 chars or ¾ of a word)
Context window	Maximum tokens the AI can process in one session
Temperature	Controls creativity/randomness (0 = precise, 1 = creative)
System prompt	Background instruction given before the conversation
Iteration	Refining a prompt based on the output received
Hallucination	When AI produces confident but incorrect information

2.4 Pro Tips: Foundations

- **Specificity beats cleverness. Be direct, not poetic.**
- **Treat your first prompt as a draft. Expect to iterate 2–3 times.**
- **State the audience, goal, and format in every serious prompt.**

CHAPTER 3

The R-C-I-F-S Core Framework

The most important structure in this handbook. Master this and you will consistently produce professional-quality prompts.

3.1 The Five Elements

Element	What It Is	Example
R — Role	Expert persona the AI should adopt	Act as a certified financial planner with 15 years of experience
C — Context	Background: audience, goal, constraints	Audience: first-year college students; goal: start SIP with ₹500/month
I — Instruction	The single, clear task you want	Create a 3-step beginner investing plan
F — Format	How output should look	3-column table: Step Action Expected Outcome
S — Structure	Logical order of the response	Start with summary, then plan, end with one caution

3.2 The Copy-Paste Skeleton

Role: You are a [profession] with [X] years of experience in [domain].
Context: I need this for [audience] because [goal].
Constraints: [tone / length / must-include / avoid].
Instruction: [The one clear task you want completed].
Format: [bullets / table / 150-word paragraph / numbered list].
Structure: Start with [X], then cover [Y], end with [Z].

3.3 Do's and Don'ts

■ Do This	■ Avoid This
Be specific about audience and goal	Write one vague line and hope
Give an example of what "good" looks like	Cram five unrelated tasks in one prompt
Ask for one thing at a time	Assume AI knows your context
Specify length and format	Accept the first draft as final
Iterate — refine instead of restarting	Forget to say who it's for

3.4 Full Example Prompt

Role: You are a professional content strategist with 8 years of experience writing for LinkedIn thought leaders.
Context: My audience is mid-career professionals (28-40 years old) who feel too busy to read. Goal: inspire them to start reading 10 minutes a day.
Instruction: Write a LinkedIn post about the ROI of daily reading.
Format: 150-180 words. Hook line, 3 bullet points, closing CTA question.
Structure: Bold stat or surprising hook -> 3 benefit bullets -> 1-line CTA.

CHAPTER 4

Structured Prompting

Structured prompting organises your prompt using deliberate formatting — labels, separators, sections, and clear delimiters — so the AI can parse your intent without ambiguity.

4.1 Techniques

Labelled Sections

```
ROLE: Senior UX designer
TASK: Review this app flow
INPUT: [paste the flow here]
OUTPUT FORMAT: Bullet list of issues and one-line fixes
```

Delimiters

```
Summarise the text between the triple dashes.
---
[Your long text here]
---
Keep summary under 100 words.
```

4.2 When to Use Structured Prompting

Situation	Benefit
Multi-part instructions	Clear separation prevents confusion
Pasting user content	Delimiters isolate data from instructions
Specifying output format	Labels make format requirements explicit
API / developer workflows	Machine-readable, consistent parsing

4.3 Common Mistakes

- Mixing instruction and input without clear separation
- Giving format instructions after the content without a clear label
- Using inconsistent markers across a long prompt

CHAPTER 5

Role-Based Prompting

Role-based prompting assigns the AI a specific expert persona before asking it to complete a task. The persona shapes vocabulary, tone, depth, assumptions, and framing.

5.1 Role Formula

```
Act as a [title] with [X] years of experience in [industry/domain],  
specialising in [specific area], working with [type of clients/audience].
```

5.2 Simple vs Enhanced Roles

Simple Role	Enhanced Role
Act as a doctor	Act as a general physician with 12 years of practice in urban India, experienced in explaining medical concepts to first-generation health seekers
Act as a writer	Act as a senior copywriter at a D2C brand, specialising in conversion-focused landing page copy for health and wellness products
Act as a teacher	Act as a Class 10 Maths teacher skilled at breaking down concepts for students with limited prior foundation

5.3 Persona Layering Checklist

- **Title** — What is their job?
 - **Experience** — How many years?
 - **Industry** — Which sector?
 - **Audience they serve** — Who do they work with?
 - **Communication style** — Formal? Conversational? Technical?
- *Try the same question across 3 different expert personas and compare. The contrast shows how powerfully roles shape responses.*

CHAPTER 6

Chain-of-Thought (CoT) Prompting

CoT prompting asks the AI to reason through a problem step by step before delivering a final answer — dramatically improving accuracy on complex tasks.

6.1 Activation Phrases

- "Think step by step"
- "Walk me through your reasoning"
- "Break this down into steps before answering"
- "Reason through this before giving a final answer"

6.2 Standard CoT Template

```
I need to solve: [your problem].  
Think step by step:  
1. List what we know  
2. Break the problem into parts  
3. Solve each part  
4. Give a clearly labelled FINAL ANSWER
```

6.3 Zero-Shot CoT vs Few-Shot CoT

Type	How It Works
Zero-Shot CoT	Add "Think step by step" to any prompt — no examples needed
Few-Shot CoT	Provide 1–2 worked examples with visible reasoning before the real question

6.4 Best Use Cases

- Financial calculations and budgeting
- Legal reasoning and case analysis
- Technical debugging
- Strategic decisions and trade-off comparisons
- Project planning and scheduling

CHAPTER 7

Zero-Shot Prompting

Zero-shot prompting gives the AI a task with no examples — relying entirely on the model's pre-trained knowledge and your instruction clarity.

7.1 When It Works Best

- Clear, well-defined tasks (summarise, translate, explain, list)
- Tasks where quality standards are flexible
- Quick first-draft generation
- Common formats (emails, blog intros, definitions)

7.2 Comparison: Basic vs Improved Zero-Shot

Version	Prompt
Basic Zero-Shot	Explain blockchain in 3 sentences for a 16-year-old.
Improved (Role + Format)	Act as a high school science teacher. Explain blockchain in exactly 3 sentences. Avoid jargon. End with one real-world analogy.

7.3 Limitations

- Style-specific outputs may be inconsistent without examples
- Niche or domain-specific tasks benefit from few-shot instead
- Complex or multi-step tasks need CoT or chaining

CHAPTER 8

Few-Shot Prompting

Few-shot prompting provides 1–3 examples of the exact input-output pattern you want, so the AI learns your style, tone, and structure and replicates it.

8.1 The Formula

```
[Optional: brief task description]

Example 1:
Input: [sample input]
Output: [desired output]

Example 2:
Input: [sample input]
Output: [desired output]

Now complete:
Input: [your actual input]
Output:
```

8.2 Example — LinkedIn Hooks

```
You write punchy LinkedIn hooks for B2B founders. Match this style:

Example 1:
Topic: Email marketing
Output: "Most founders ignore email. Their competitors don't."

Example 2:
Topic: Cold outreach
Output: "Your cold email isn't failing because of the subject line. It's the first sentence."

Now write a hook for:
Topic: Hiring your first salesperson
Output:
```

8.3 Few-Shot vs Zero-Shot Comparison

Dimension	Zero-Shot	Few-Shot
Examples needed	None	1–3
Style control	Low	High
Best for	Common formats	Niche/style-specific outputs
Sweet spot	—	2 examples

■ 2 examples is usually the sweet spot. One may not generalise; three or more can over-constrain. Always use your best work as examples.

CHAPTER 9

XML Prompting

XML prompting uses structured XML tags to clearly separate instructions, data, context, and output specifications with machine-readable precision. Heavily recommended for API and agentic workflows, and especially effective with Claude.

9.1 Basic Structure

```
Summarise the customer review below

Product review for an e-commerce platform.
Audience: operations team reviewing product quality.

[Paste the customer review here]

- Sentiment: Positive / Negative / Neutral
- Key Issue (if any): one sentence
- Suggested Action: one sentence
```

9.2 When to Use XML

Use Case	Why XML Helps
Document analysis pipelines	Clearly separates document from instruction
Multi-step workflows	Each step can be tagged and referenced
Extracting structured data	Output tags enable reliable parsing
System prompts for products	Clean separation of persona, rules, and task

9.3 Claude-Specific Advantage

Claude responds exceptionally well to XML-structured prompts. When building production workflows or system prompts, XML is the preferred format for clarity and reliability.

CHAPTER 10

Context Engineering

Context engineering is the practice of managing what information the AI has access to — including what to include, what to exclude, how to compress it, and when to reset it.

10.1 Context Window Reference

Model Type	Approximate Context Window
Standard models	8,000 – 32,000 tokens
Extended models	100,000 – 200,000 tokens
Claude 3+	Up to 200,000 tokens

10.2 The Memory Block Technique

When a conversation grows long, summarise key context into a compact block and re-paste it at the start of a new session:

```
MEMORY BLOCK (paste at start of each new session):  
- Project: Redesign of loyalty program  
- My role: UX Consultant  
- Client constraints: Budget under Rs.10L, launch by March  
- Decisions made: Tier system confirmed (Silver/Gold/Platinum)  
- Last task completed: User journey map draft  
- Next task: Wireframe for onboarding flow
```

10.3 The Caveman Trick (Context Compression)

Normal	Compressed (Caveman)
We have decided that the app will target urban women aged 22–35 who are interested in sustainable fashion and income.	Target: urban women, 22-35, sustainable fashion, high income.

10.4 Fresh vs Continue Decision

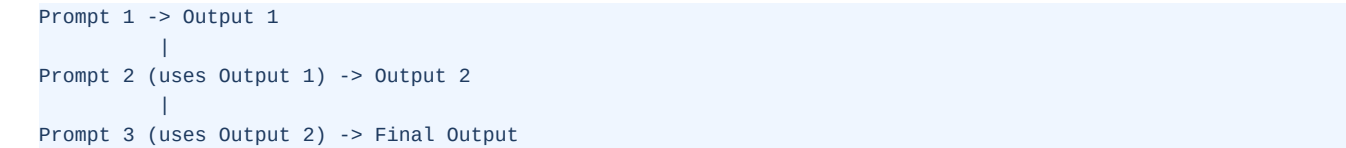
Situation	Recommendation
New unrelated task	Start fresh conversation
Continuing same project	Keep thread, add memory block if long
AI seems confused or off-track	Fresh session with clean compressed summary
Output quality degrading	Fresh session with better initial prompt

CHAPTER 11

Prompt Chaining

Prompt chaining breaks a complex task into a sequence of smaller, focused prompts — where the output of one prompt becomes the input to the next.

11.1 Chain Design Pattern



11.2 Example — Blog Post Chain

Step	Prompt
1 — Research	Act as a content strategist. List 7 data-backed reasons why sleep deprivation harms productivity. One statistic per point. Bullet list.
2 — Outline	Using these 7 points [paste Output 1], create a structured outline for a 1,000-word blog post targeting stressed corporate professionals.
3 — Draft	Using this outline [paste Output 2], write the full 1,000-word blog post. Tone: conversational and empathetic.
4 — Polish	Review this blog post [paste Output 3]: 1. Improve headline 2. Tighten paragraphs 3. Strengthen CTA.

11.3 Use Cases

- Long-form content (books, reports, courses)
- Research → analysis → recommendation workflows
- Data cleaning → analysis → visualisation scripts
- Job application (research → tailor CV → write cover letter)
- Product development (ideation → user stories → feature specs)

CHAPTER 12

AI Agents & Agentic Prompting

An AI agent takes autonomous sequences of actions — using tools, making decisions, and operating across multiple steps — to complete a complex goal with minimal human intervention.

12.1 Key Concepts

Concept	Description
Tool use	AI can call external functions (web search, code execution, APIs)
Memory	Persistent context across sessions (short-term or long-term)
Planning	AI breaks a goal into sub-tasks autonomously
Action loops	AI takes action, observes result, takes next action
Human-in-the-loop	Points where human review or approval is required

12.2 System Prompt Template for Agents

```
You are a research assistant agent.  
Goal: answer questions by searching, synthesising, and citing sources.  
  
TOOLS: web_search, document_reader, calculator  
  
RULES:  
- Verify claims using at least 2 sources  
- Ask for clarification before irreversible actions  
- Summarise your plan before executing  
- Report errors immediately  
  
OUTPUT: 1. Action taken 2. Finding 3. Source 4. Confidence: High/Medium/Low
```

12.3 Safety Principles

- **Minimal footprint:** Request only necessary permissions
- **Confirm before irreversible actions:** File deletion, sending emails, payments
- **Pause points:** Build in check-ins at key decision points
- **Clear error handling:** Tell the agent what to do with unexpected input

CHAPTER 13

Claude Best Practices

13.1 What Makes Claude Different

Feature	Detail
Instruction-following	Strong accuracy across complex, multi-part tasks
Context window	Up to 200K tokens — ideal for long document analysis
Nuanced reasoning	High accuracy on analytical and sensitive tasks
Structured prompts	Excels with XML and labelled-section formats
Careful responses	Thoughtful handling of ambiguous or sensitive areas

13.2 Claude-Specific Tips

Tip	How To Apply
Use XML tags	Claude responds exceptionally well to XML-structured prompts
Be explicit about thinking	Use: "Think carefully before answering" or "Work through this step by step"
Leverage system prompt	Put all persona, rules, constraints in system prompt via API — keep user messages clean
Long-context strength	Feed entire contracts, reports, or transcripts for deep analysis
Self-critique loop	Add: "Review your answer. What is the weakest part? How would you improve it?"

13.3 Claude System Prompt Template (API)

```
# SYSTEM PROMPT
You are [role] with [experience].
Your communication style is [tone].
Always [key rule 1] and [key rule 2].
Never [key restriction].

# USER MESSAGE
[Clear single instruction]
[Background, audience, goal]
[User content or question]
[Exact format required]
```